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Scenario 1: Student Academic Performance Dashboard

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
data = {
    "Student ID":
["S001","S002","S003","S004","S005","S006","S007","S008","S009","S010"],
    "Attendance (%)": [95,88,76,82,91,68,79,97,84,73],
    "Internal Marks": [92,85,72,78,90,65,75,96,81,70],
    "Department": ["CSE","AI","IT","ECE","CSE","AI","IT","ECE","CSE","AI"]
}
```

```
df = pd.DataFrame(data)
```

```
plt.figure(figsize=(8,6))
```

```
sns.scatterplot(data=df,
```

```
x="Attendance (%)",
```

```
y="Internal Marks",
```

```
hue="Department",
```

```
palette="Set2",
```

```
s=120)
```

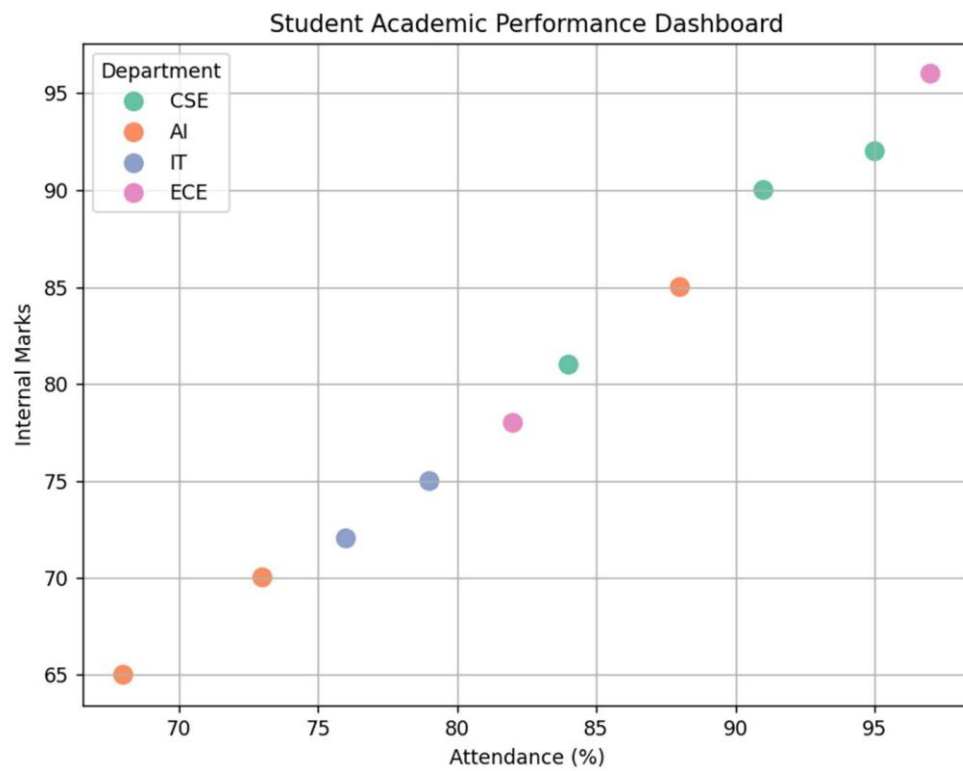
```
plt.title("Student Academic Performance
```

```
Dashboard") plt.xlabel("Attendance (%)")
```

```
plt.ylabel("Internal Marks")
```

```
plt.legend(title="Department") plt.grid(True)
```

```
plt.show()
```



Scenario 2: Electric Vehicle Market Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
data = {
    "Vehicle Model":["EV1","EV2","EV3","EV4","EV5","EV6","EV7","EV8"],
    "Battery Capacity":[30,40,45,50,55,60,70,80],
    "Price":[10,13,15,18,20,23,28,32],
    "Driving Range":[250,300,340,380,420,460,520,600],

    "Manufacturer":["Tata","MG","Mahindra","Hyundai","Tata","MG","Mahindra","Hyundai"]
}
```

```
df=pd.DataFrame(data)
```

```
colors={"Tata":"blue","MG":"green","Mahindra":"red","Hyundai":"orange"}
```

```
plt.figure(figsize=(8,6))
```

```
for m in df["Manufacturer"].unique():
```

```
    d=df[df["Manufacturer"]==m]
```

```
    plt.scatter(d["Battery Capacity"],
```

```
               d["Price"],
```

```
               s=d["Driving Range"]/2,
```

```
               color=colors[m],
```

```
               alpha=0.7,          label=m)
```

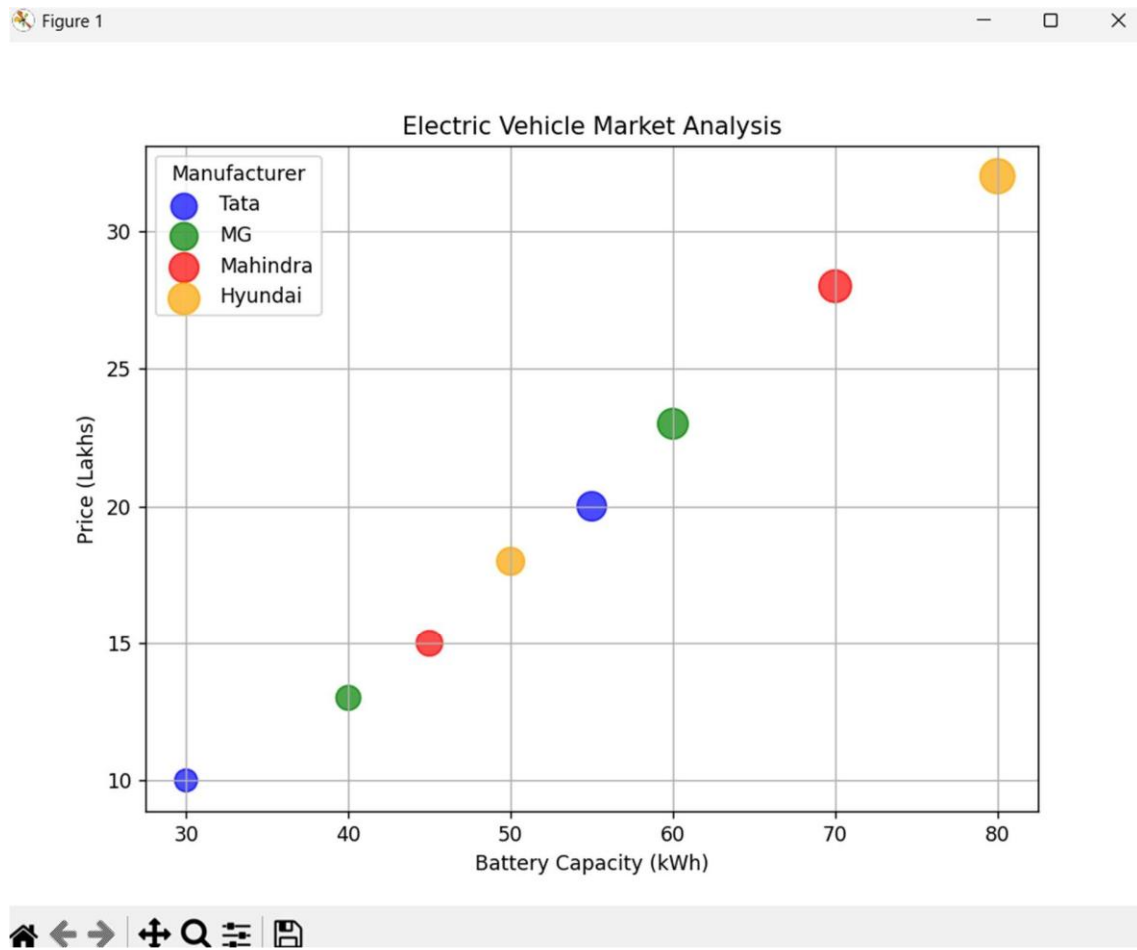
```
plt.title("Electric Vehicle Market
```

```
Analysis") plt.xlabel("Battery Capacity
```

```
(kWh)") plt.ylabel("Price (Lakhs)")
```

```
plt.legend(title="Manufacturer")
```

```
plt.grid(True) plt.show()
```



Scenario 3: Hospital Patient Admission Analysis

```
import pandas as pd
import matplotlib.pyplot as plt

data={
    "Department":["Cardiology","Neurology","Orthopedics","Pediatrics","General"],
    "Number of Patients":[120,80,95,150,200],
    "Average Treatment Cost":[45000,60000,35000,20000,15000]
}

df=pd.DataFrame(data)

colors=["red","blue","green","orange","purple"]

plt.figure(figsize=(8,6))

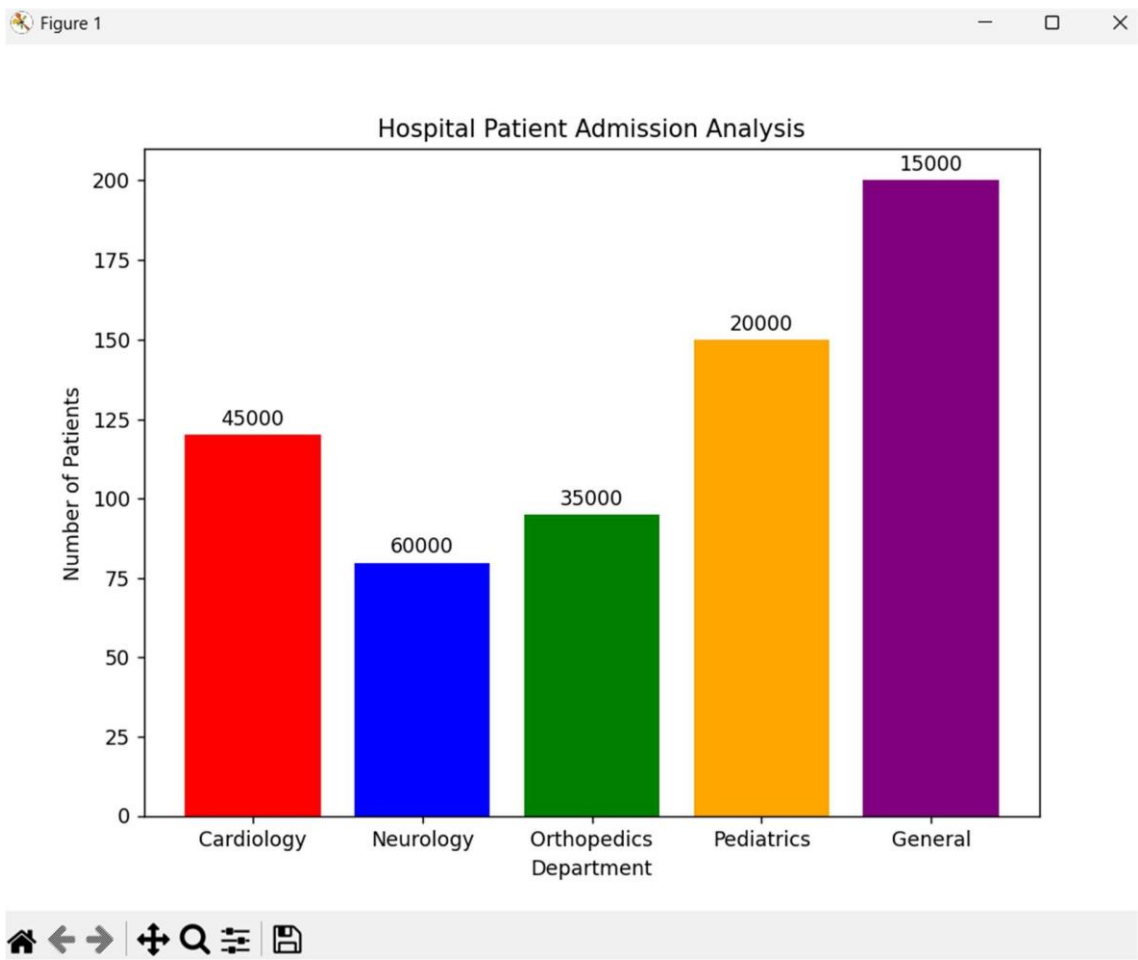
bars=plt.bar(df["Department"],
df["Number of Patients"],
color=colors)

for bar,cost in zip(bars,df["Average Treatment Cost"]):

    plt.text(bar.get_x()+bar.get_width()/2,
bar.get_height()+3,      str(cost),
ha="center")

plt.title("Hospital Patient Admission
Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")
```

plt.show()



Scenario 4: Weather Monitoring System

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
data={  
    "Date":[1,2,3,4,5,6,7],  
    "Chennai":[34,35,36,35,34,33,35],  
    "Bangalore":[27,28,29,28,27,26,27],  
    "Delhi":[37,38,39,40,39,38,37]  
}
```

```
df=pd.DataFrame(data)
```

```
plt.figure(figsize=(8,6))
```

```
plt.plot(df["Date"],  
df["Chennai"],  
color="red",      linestyle="-",  
marker="o",  
label="Chennai")
```

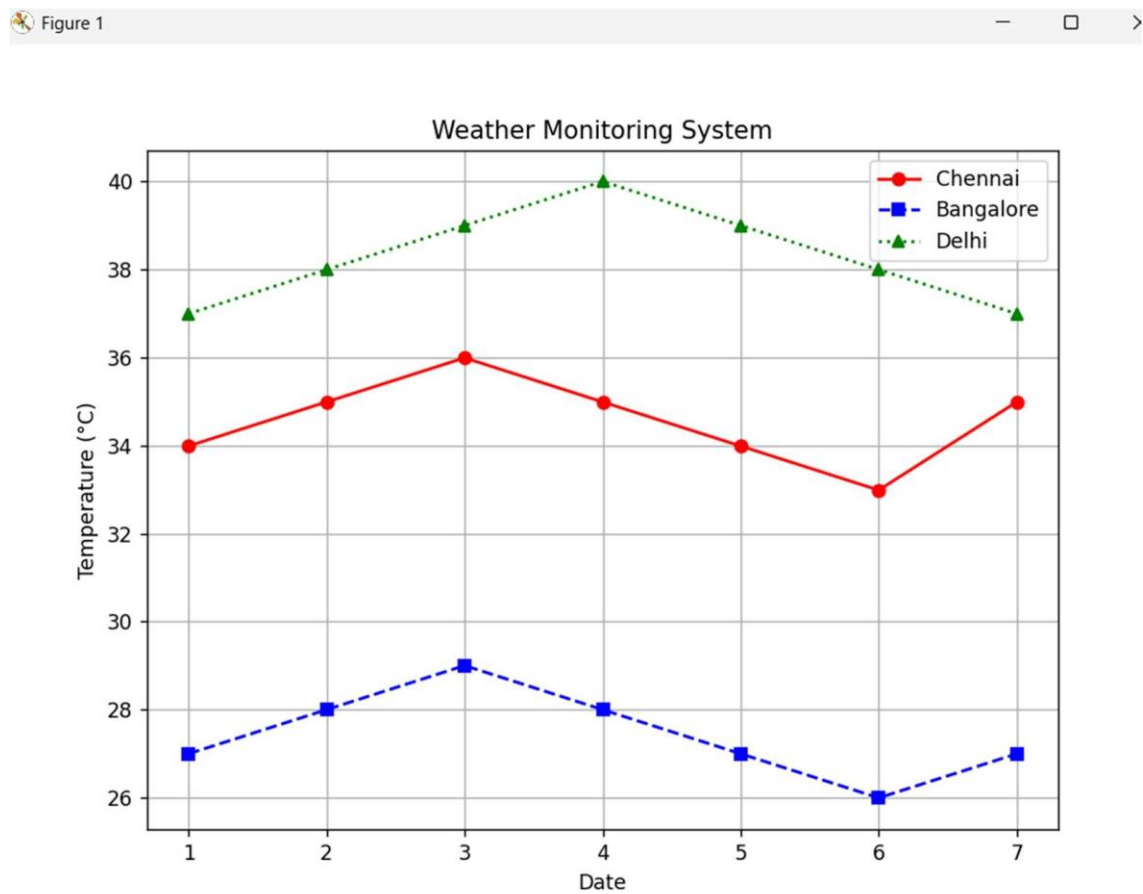
```
plt.plot(df["Date"],  
df["Bangalore"],  
color="blue",     linestyle="-",  
marker="s",  
label="Bangalore")
```

```
plt.plot(df["Date"],  
df["Delhi"],  
color="green",
```

```
linestyle=":",  
marker="^",  
label="Delhi")
```

```
plt.title("Weather Monitoring  
System") plt.xlabel("Date")  
plt.ylabel("Temperature (°C)")  
plt.legend() plt.grid(True)
```

```
plt.show()
```



Scenario 5: Supermarket Sales Analysis

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn
```

```
as sns
```

```
data={  
    "Month":["January", "January", "January",  
            "February", "February", "February",  
            "March", "March", "March"],  
    "Product Category":["Groceries", "Beverages", "Snacks"]*3,  
    "Sales Amount":[50000, 30000, 25000,  
                    55000, 32000, 27000,  
                    60000, 35000, 30000],  
    "Branch":["North", "South", "East",  
             "North", "South", "East",  
             "North", "South", "East"]  
}
```

```
df=pd.DataFrame(data)
```

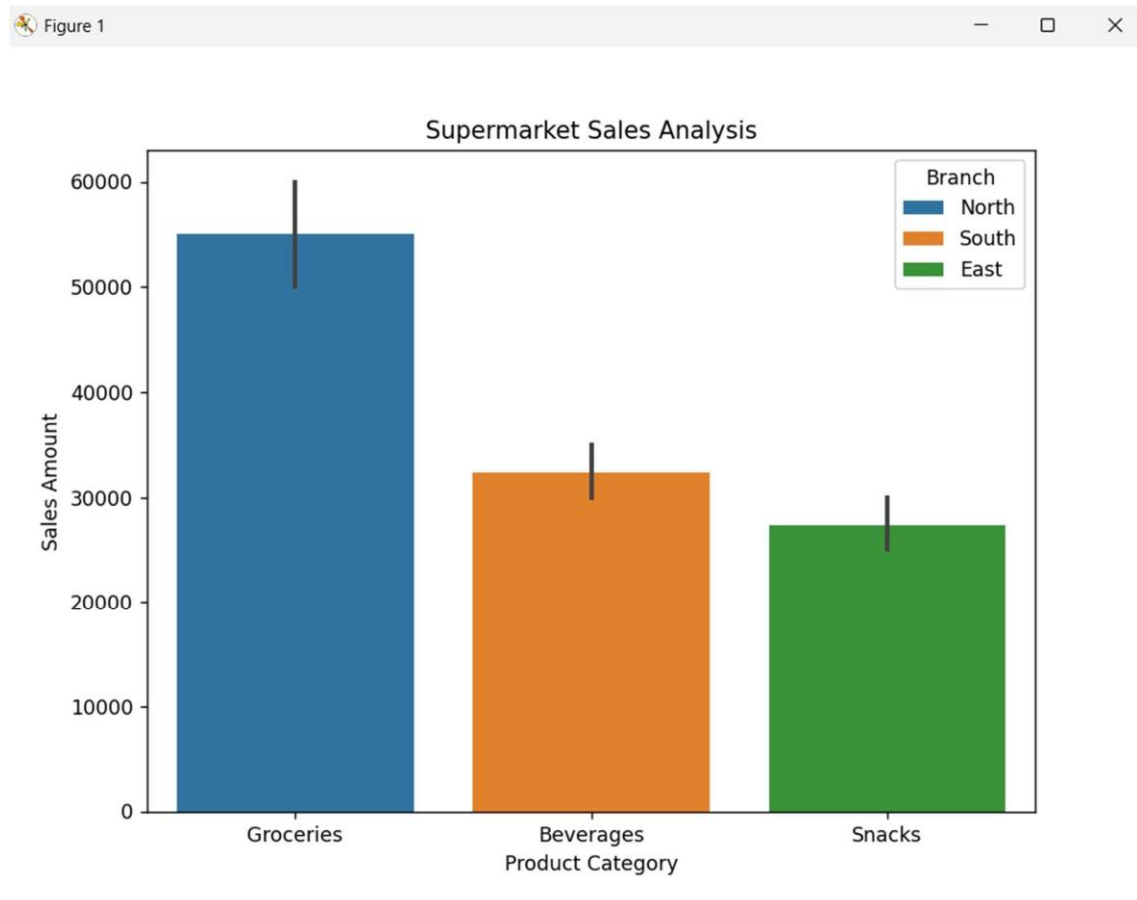
```
plt.figure(figsize=(8,6))
```

```
sns.barplot(data=df,  
            x="Product Category",  
            y="Sales Amount",  
            hue="Branch")
```

```
plt.title("Supermarket Sales  
Analysis")  
plt.xlabel("Product
```

```
Category") plt.ylabel("Sales  
Amount") plt.legend(title="Branch")
```

```
plt.show()
```



Scenario 6: Employee Salary Distribution

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
data = {
```

```
    "Employee ID":
```

```
    ["E001", "E002", "E003", "E004", "E005", "E006", "E007", "E008", "E009", "E010"],
```

```
    "Salary": [25000, 30000, 35000, 40000, 45000, 50000, 55000, 60000, 65000, 70000],
```

```
    "Experience": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
```

```
    "Department":
```

```
    ["HR", "IT", "Finance", "Marketing", "HR", "IT", "Finance", "Marketing", "HR", "IT"]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
plt.figure(figsize=(8,6))
```

```
plt.hist(df["Salary"],
```

```
    bins=5,
```

```
    color="skyblue",
```

```
    edgecolor="black")
```

```
plt.title("Employee Salary
```

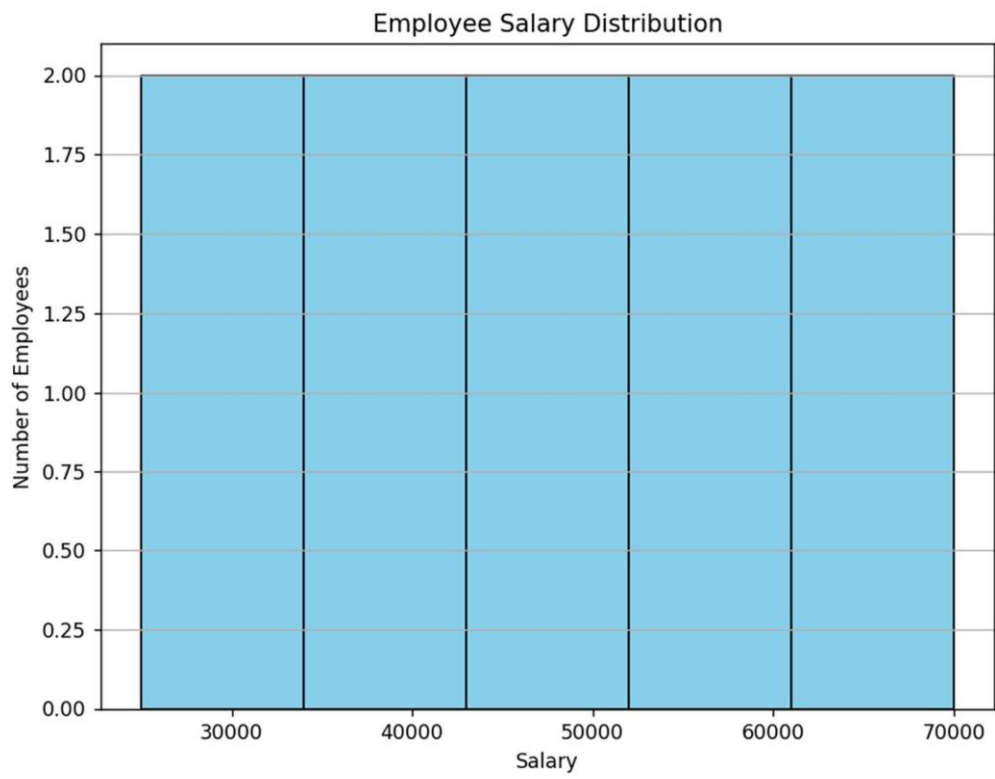
```
Distribution") plt.xlabel("Salary")
```

```
plt.ylabel("Number of Employees")
```

```
plt.grid(axis="y")
```

```
plt.show()
```

Figure 1



Scenario 7: Smartphone Market Comparison

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
data = {
    "Brand":
["Samsung","Apple","OnePlus","Xiaomi","Samsung","Apple","OnePlus","Xiaomi"],
    "RAM": [6,8,12,8,16,12,8,6],
    "Processor Speed": [2.4,3.2,3.0,2.8,3.3,3.4,3.1,2.7],
    "Price": [25000,80000,45000,22000,90000,100000,50000,20000],
    "Storage": [128,256,256,128,512,512,256,64]
}
```

```
df = pd.DataFrame(data)
```

```
markers = {
    "Samsung": "o",
    "Apple": "s",
    "OnePlus": "^",
    "Xiaomi": "D"
}
```

```
plt.figure(figsize=(8,6))
```

```
for brand in df["Brand"].unique():
    temp = df[df["Brand"] == brand]
```

```
plt.scatter(temp["RAM"],
```

```

temp["Price"],
s=temp["Storage"],
c=temp["Processor Speed"],
cmap="viridis",
marker=markers[brand],
label=brand)

```

```

plt.colorbar(label="Processor Speed
(GHz)") plt.title("Smartphone Market
Comparison") plt.xlabel("RAM (GB)")
plt.ylabel("Price (₹)")
plt.legend(title="Brand") plt.grid(True)

```

```

plt.show() import pandas as
pd import matplotlib.pyplot
as plt

```

```

# Dataset

```

```

data = {
    "Brand":
["Samsung","Apple","OnePlus","Xiaomi","Samsung","Apple","OnePlus","Xiaomi"],
    "RAM": [6,8,12,8,16,12,8,6],
    "Processor Speed": [2.4,3.2,3.0,2.8,3.3,3.4,3.1,2.7],
    "Price": [25000,80000,45000,22000,90000,100000,50000,20000],
    "Storage": [128,256,256,128,512,512,256,64]
}

```

```

df = pd.DataFrame(data)

```

```

markers = {

```

```
"Samsung": "o",  
"Apple": "s",  
"OnePlus": "^",  
"Xiaomi": "D"  
}
```

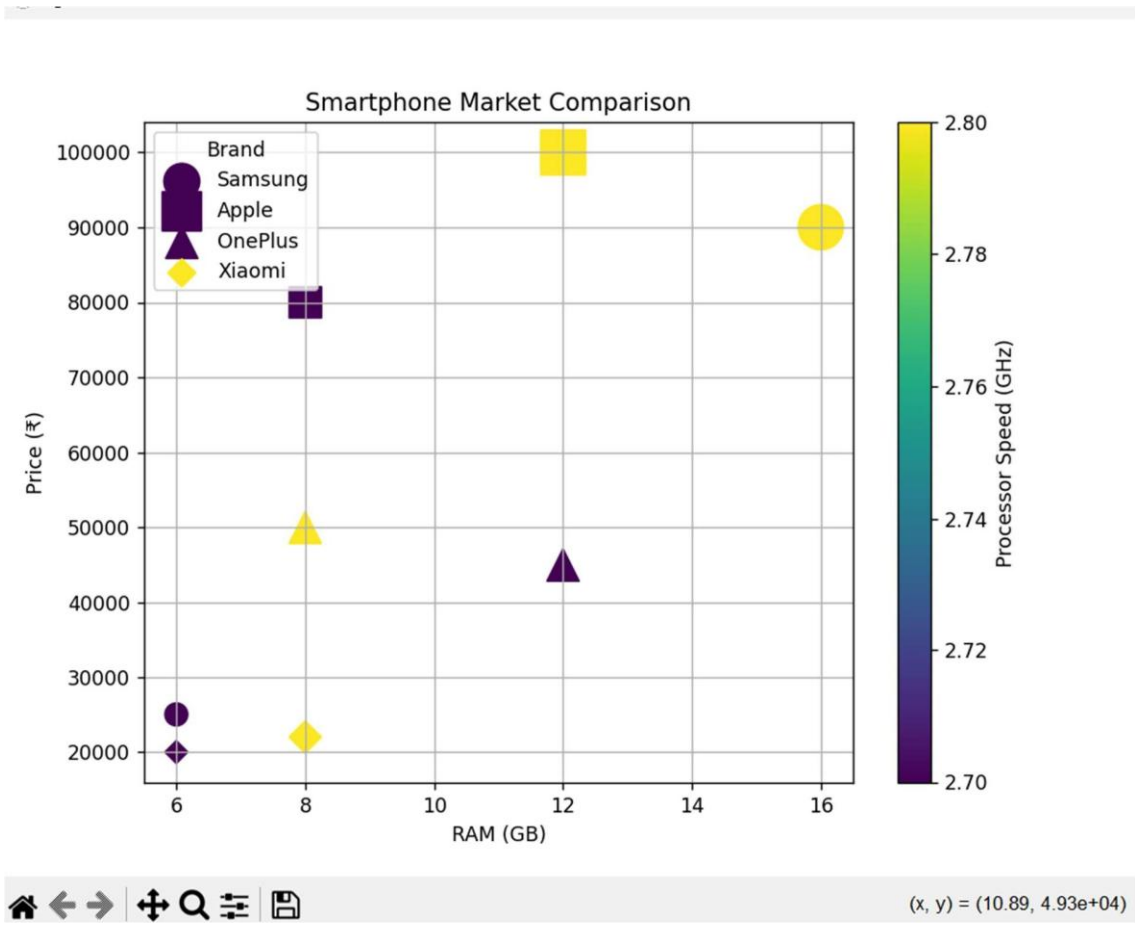
```
plt.figure(figsize=(8,6))
```

```
for brand in df["Brand"].unique():  
    temp = df[df["Brand"] == brand]
```

```
    plt.scatter(temp["RAM"],  
                temp["Price"],  
                s=temp["Storage"],  
                c=temp["Processor Speed"],  
                cmap="viridis",  
                marker=markers[brand],  
                label=brand)
```

```
plt.colorbar(label="Processor Speed  
(GHz)") plt.title("Smartphone Market  
Comparison") plt.xlabel("RAM (GB)")  
plt.ylabel("Price (₹)")  
plt.legend(title="Brand") plt.grid(True)
```

```
plt.show()
```



scenario 8 online movie rating analysis

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn
```

```
# Dataset
```

```
data = {
```

```
    "Genre": ["Action", "Comedy", "Drama", "Horror", "Sci-Fi"],
```

```
    "2022": [7.5, 6.8, 8.2, 6.5, 8.0],
```

```
    "2023": [7.8, 7.0, 8.5, 6.9, 8.3],
```

```
    "2024": [8.0, 7.2, 8.7, 7.1, 8.6]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
heatmap_data = df.set_index("Genre")
```

```
plt.figure(figsize=(8,5))
```

```
sns.heatmap(heatmap_data,
```

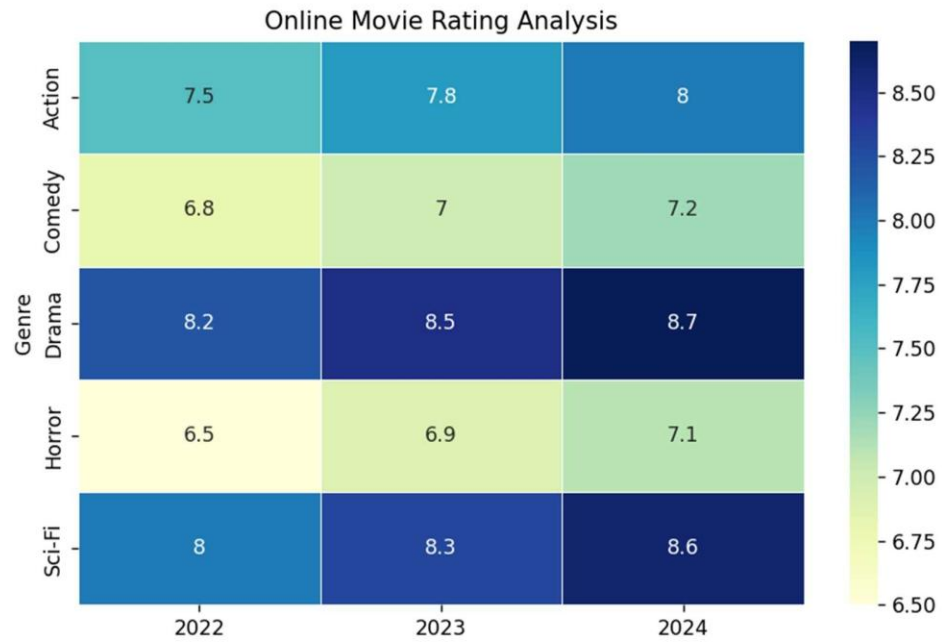
```
            annot=True,
```

```
            cmap="YlGnBu",
```

```
            linewidths=0.5)
```

```
plt.title("Online Movie Rating Analysis")
```

```
plt.show()
```



Scenario 9: COVID-19 State-wise Analysis (Tile Plot) import

pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

Dataset

data = {

 "State": ["Tamil Nadu", "Kerala", "Karnataka", "Maharashtra", "Delhi", "Telangana"],

 "Active Cases": [2500, 1800, 2100, 4500, 1200, 1700],

 "Vaccination Percentage": [92, 95, 91, 89, 94, 90],

 "Population": [72000000, 35000000, 68000000, 124000000, 19000000, 39000000]

}

df = pd.DataFrame(data)

tile_data = df.set_index("State")[["Active Cases"]]

plt.figure(figsize=(6,5))

sns.heatmap(tile_data,

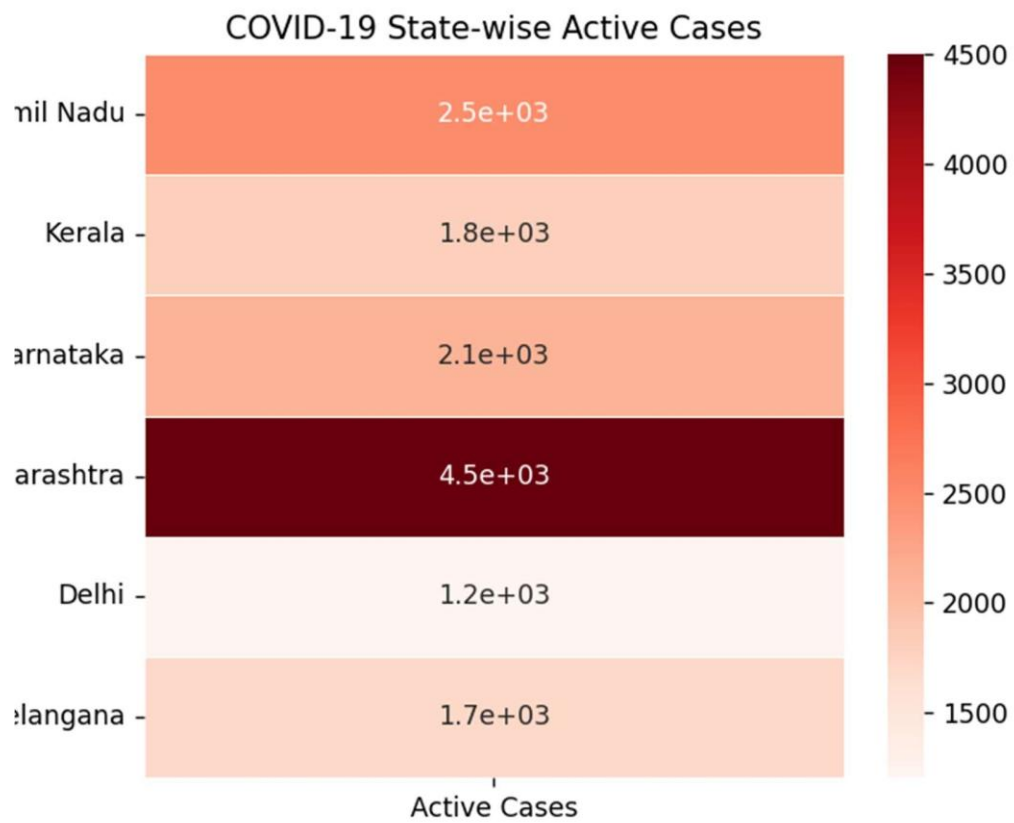
annot=True,

cmap="Reds",

linewidths=0.5)

plt.title("COVID-19 State-wise Active Cases")

plt.show()



Scenario 10: Iris Flower Classification

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
data = {
```

```
    "Sepal Length": [5.1, 4.9, 5.8, 6.0, 6.5, 6.3, 5.0, 6.7, 6.9],
```

```
    "Sepal Width": [3.5, 3.0, 2.7, 2.9, 3.0, 3.3, 3.4, 3.1, 3.2],
```

```
    "Petal Length": [1.4, 1.4, 5.1, 4.5, 5.8, 6.0, 1.5, 4.7, 5.4],
```

```
    "Petal Width": [0.2, 0.2, 1.9, 1.5, 2.2, 2.5, 0.2, 1.4, 2.1],
```

```
    "Species": ["Setosa", "Setosa", "Virginica", "Versicolor", "Virginica", "Virginica", "Setosa", "Versicolor", "Virginica"]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
colors = {
```

```
    "Setosa": "red",
```

```
    "Versicolor": "green",
```

```
    "Virginica": "blue"
```

```
}
```

```
markers = {
```

```
    "Setosa": "o",
```

```
    "Versicolor": "s",
```

```
    "Virginica": "^"
```

```
}
```

```
plt.figure(figsize=(8,6))
```

```
for species in df["Species"].unique():
```

```
    temp = df[df["Species"] == species]
```

```
    plt.scatter(temp["Sepal Length"],  
temp["Petal Length"],  
s=temp["Sepal Width"]*80,  
color=colors[species],  
marker=markers[species],  
label=species)
```

```
plt.title("Iris Flower Classification")
```

```
plt.xlabel("Sepal Length (cm)")
```

```
plt.ylabel("Petal Length (cm)")
```

```
plt.legend(title="Species")
```

```
plt.grid(True)
```

```
plt.show()
```

